1. Cooling system and how a pressure sensor is used

Basic Physics of Cooling Battery and Pressure:
Bernoulli's Principle: Total energy = Static Pressure + density*velocity^2
P + 1/2pv^2 + pgh = constant

(If the kinetic energy of the air increases, the potential energy of the static pressure must drop)

PV = nRT

Assuming an air-cooled accumulation system: an accumulator has a sensor to measure the pressure drop (Pa)

- We need one sensor at the inlet, and another sensor at the outlet → **differential pressure sensor**

Steady-state airflow: stable, continuous airflow when cooling fans run at a constant speed and the vehicle drives at a uniform velocity

- Differential pressure (delta p) ~ constant/shifts slowly over minutes
- Sampling speed of 1-5Hz

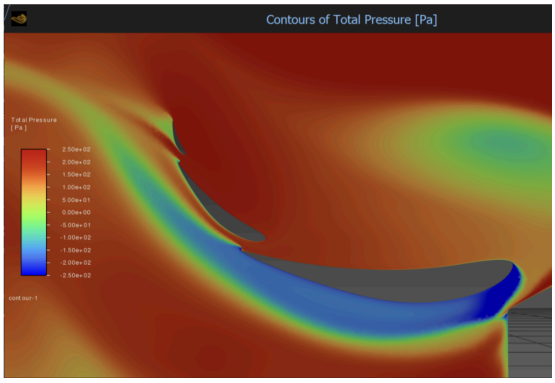
Transient Airflow: Rapidly changing airflow caused by stepping on the throttle or sudden vehicle aerodynamic pressure shifts

- The pressure drop spikes/decreases in milliseconds
- Sampling rate of 100 Hz to 500 Hz

2. Pressure sensor Specifications that are important for this application:

1. **Range**: the smallest to largest value the sensor is capable of measuring.

2. **Resolution**: the smallest change is that the sensor can distinguis/report
Ex. 10005 Pa vs 10004 Pa if the sensor has a resolution of 1 Pa
3. **Accuracy**
4. **Precision**: if I measure repeatedly, how often do I get the same result?



Based on this graph, the entire aerodynamic phenomenon I am trying to measure is just 500 Pa wide.

- Atmospheric pressure is 101,325 Pa

3. Different Types of Sensors

Absolute vs. differential pressure sensors:
Absolute:

Differential pressure sensor:
could be used for things like intake restrictor monitoring (you have Port A and Port B; do we bring these 2 ports together?)

- **Digital Barometric Array:** network cluster of multiple MEMS either on a circuit board or in vehicle chassis.

- **MEMS**: its mechanical pressure diaphragm is down to the micrometer level. High pressure pushes on the silicon diaphragm, and lower air pressure pushes up against the bottom. P1 > P2 → the diaphragm bends down a few nanometers → ASIC passes current through resistors and measures the change in voltage bc of the bending → converts it to Pa through a built-in calibration equation.
    - Extremely lightweight, take up a few square milimeters, measures pressure changes down to 0.1 Pa. low power
    - Digital MEMS sensors often support I2C/SPI. I^2C allows multiple devices to share SDA/SCL only if each can use a unique address. Fixed-address sensors require separate I2C buses. SPI avoids address collisions by giving each sensor its own chip-select line.

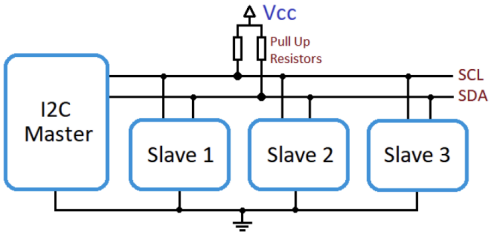
4. Communication Interfaces for Sensors

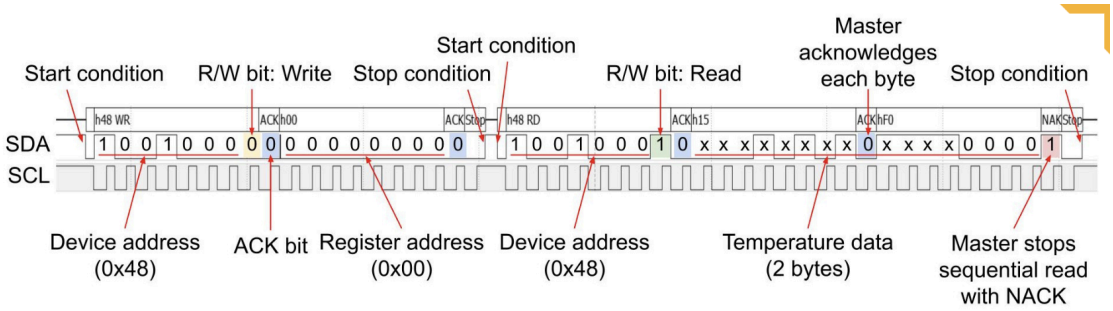
Types of Communication interfaces and trade-offs

Type of Interface	Data rate	Distance	Power / Pins	Notes
SPI	25 Mbps high	~20cm, Short best for comm on the same PCB or nearby boards	4 shared signal lines (SCLK, MOSI, MISO, CS)	fast and good for multiple identical sensors because each can have its own pin.
I2C	1Mbps, medium	~1m, same PCB or short wiring	2 shared signal pins(SDA and SCL)	many devices can share the same bus, but every device needs a unique address.
Onewire	low	longer wiring than SPI/I2C, depd on layout	Usually 1 data pin plus ground	each device has a unique ID, so many devices can share *one wire* Not used for high-speed sensors.
Analog	no fixed digital data rate	Short to moderate, but longer wires are susceptible to noise	Usually 1 analog input per sensor	sensor outputs a voltage/current requires an ADC
UART	25kbps, low	~15m, longer	2 signal pins	1 transmitter talks

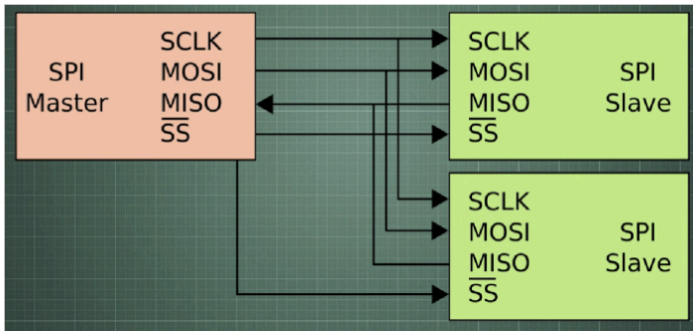
		than PCB buses if designed carefully	(TX and RX)	to 1 receiver →not designed for identical sensors sharing 1 bus

I2C:

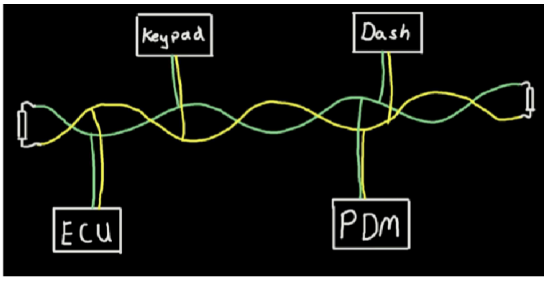




SPI:

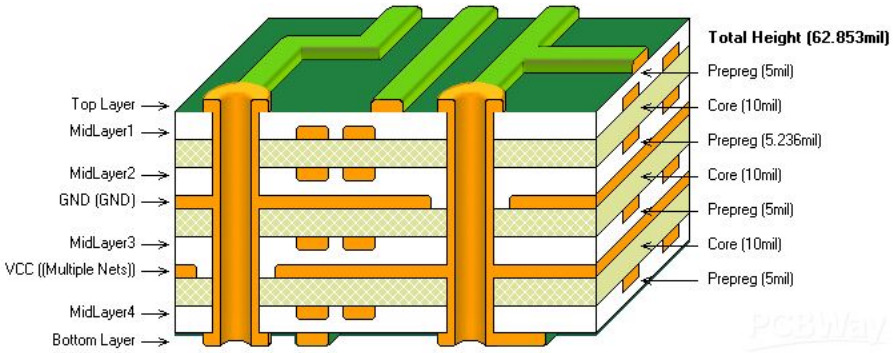


CAN:



PCB communication through different protocols like SPI, I2C. Physically - multiple layers in PCB to route signals (and power)

- intra-PCB communication happens in the layers of the board:



5. Sensor Requirements

- ☐ Differential Sensor: we need to calculate the difference between the inlet and outlet to find out the change in pressure
- ☐ Resolution: <= 1 Pa (similar or better than BMP180)
- ☐ Range: min of 500 Pa
- ☐ Interface: SPI preferred, then I2C

What accuracy do we need? Around 50 Pa to 10 Pa.
What sample rate?

- Are we testing steady-state or transient airflow? Steady rate. Transient is way too complicated.

What is the temperature range of the battery? 45C to 50C

We are measuring changes in temperature and change in pressure (Pin and Pout)

Bosch BMP390